G.A.T.E.S. · PROFESSOR SUBMISSION

Gate 1 — final validation report

Combined controlled-run analysis, check audit, feedback-loop evidence, generated-paper verification, MLR-Bench context, and completion decision.

16 August 2026 · config a0409cf020f3… · 10 gated + 10 ungated artifacts · DeepSeek V4 Flash

This report reconciles the **Claude quantitative report** and the **Codex controlled validation report** against their shared retained evidence. Every headline number is re-derived at build time.

00 Executive conclusion

40/40	96.6%	9/18	362
required values delivered to the writer	gated paper claims traceable to the run	rejected turns the legacy rule could not see	regression tests passing at campaign close
legacy view of the same artifacts: 0/40	direct registry matches + checkable derivations	three deterministic repetitions; zero mismatches	286 Gates + 76 Agent Laboratory

Completion decision: accept Gate 1 for its declared scope.

Gate 1 establishes deterministic execution validity, typed result capture, complete-log retention, and source-to-value provenance. It materially improved evidence delivery and paper traceability. It did not establish a paper-quality effect and does not judge whether a method or scientific interpretation is sound.

The strongest result is paired and mechanical: the same ten gated executions deliver 40/40 required values through Gate 1 and 0/40 through the reconstructed legacy 1,000-character channel.

01 Purpose and trust boundary

Agent Laboratory previously treated the first 1,000 characters of stdout as its only experiment-evidence channel. Gate 1 inserts a deterministic boundary between agent-written code and downstream claims so that an execution must complete validly and expose typed, traceable results before the writing phase can cite them.

STAGE	GATE 1 ACTION	ARTIFACT RETAINED

Candidate source	Compile and statically inspect	source plus SHA-256 identity
Execution	Run in a bounded child process	argv, duration, exit, exception, environment
Result contract	Capture typed keys and finite values	results.json and citable registry
Decision	FAIL blocks; WARN/INFO diagnose	gate1_report.json and divergence ledger
Writer hand-off	Place registry before any stdout budget	evidence bundle with full-log paths

Verdict invariant. Only deterministic checks can block. The optional model-assisted log scan and generated repair prose remain diagnostic and cannot change PASS or FAIL.

02 Controlled testing ground

Both completed arms used the same secret-free YAML, task, prompts, model role configuration, and four-key result contract. The treatment was the environment switch GATES_GATE1=on versus off.

PARAMETER	GATE 1 ON	GATE 1 OFF	CONTROL
Configuration SHA-256	`a0409cf020f37a9c…`	`a0409cf020f37a9c…`	identical
Model	DeepSeek V4 Flash	DeepSeek V4 Flash	same
Completed artifacts	10	10	same count
Required keys	4	4	same contract
Final paper	produced	produced on completed retry	both retained in this bundle
Wall-clock	61.2 min	146.2 min	ungated retry ran serially

> *“Report a numeric value only when that value was supplied by the experiment evidence available to the writing phase. If a required value is absent, state that it is unavailable; do not estimate or reconstruct it.”*

The task deliberately prints more than 1,000 characters of training logs before its final measurements. This turns the legacy truncation boundary into a controlled stressor rather than an incidental detail.

03 Feedback loop and retained report

A failed engineer turn receives a bounded, structured report: exact check IDs, source location, full-log paths, execution provenance, and grounded repair guidance. The next rewrite is evaluated from scratch; after three rejected turns the loop stops without handing invalid evidence to the paper writer.

LOOP STEP	BEHAVIOR
1. Submit	Engineer supplies candidate experiment source
2. Evaluate	Static, execution, environment, result, and log checks run
3. Report	FAIL returns precise evidence; WARN/INFO remain visible
4. Repair	Next engineer turn receives the report and full-log references
5. Stop	PASS advances; three rejected turns raise GateFailure

```
GATE 1 — EXECUTION VALIDITY: FAIL (attempt 2 of 3)
[exec.exit_code_zero] exited 1
[exec.no_uncaught_exception] NameError: name 'correct' is not defined
[results.contract_present] no results were declared — the run crashed before any record_result() call
completed
[env.seed_recorded] WARN — no seed was declared
[output.untruncated] 3,420 bytes of stdout captured in full
source line 8: test_acc = correct / total
```

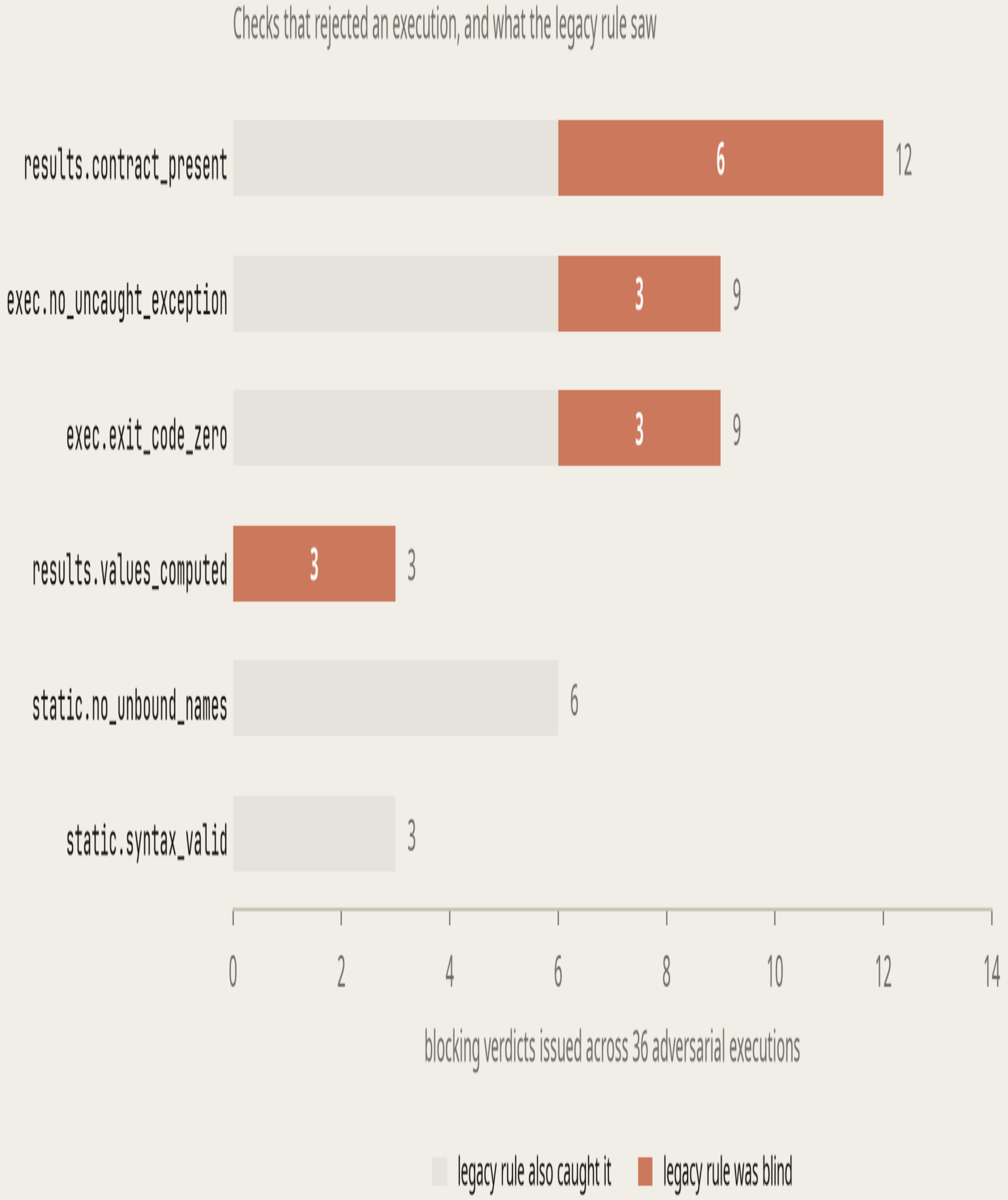
THREE DETERMINISTIC REPETITIONS	COUNT
Scenario runs	15
Executions	36
Rejected engineer turns	18
Rejected turns legacy would accept	9
Expectation mismatches	0

The retained example is bundled at verification/evidence/feedback_loop_example/. Its NameError marker appears after the legacy 1,000-character window, so the older rule would have accepted the failed run.

04

All checks and observed attribution

Gate 1 declares 22 checks: 13 blocking, six warning-tier, and three informational. All 13 blockers passed in all ten gated full-workflow executions. The adversarial corpus supplies the failure attribution that an all-pass campaign cannot.



Blocking verdicts by check across 36 adversarial executions. Coral marks failures the legacy detector could not see.

9 of 18 rejected turns (50%) were invisible to the legacy rule. At execution level, 9 of 24 failed executions (37.5%) were legacy-blind. The turn-level 9/18 figure is the feedback-loop headline; 9/24 is the separate check-execution unit.

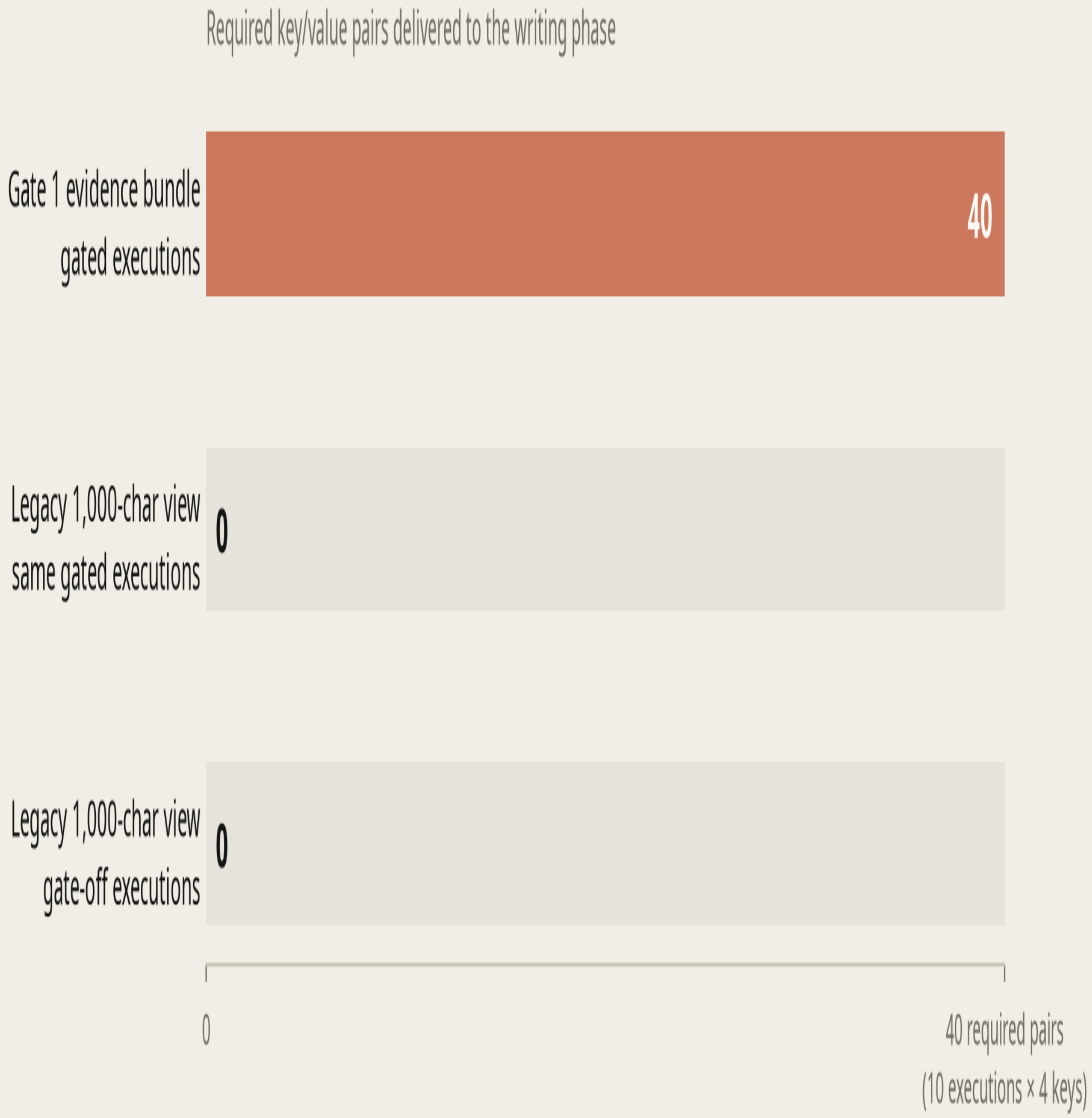
CHECK ID	TIER	WHAT IT ESTABLISHES	GATED CAMPAIGN	CORPUS FIRES
`results.contract_present`	FAIL	at least one typed result exists	10/10	12
`exec.exit_code_zero`	FAIL	the child exited zero	10/10	9
`exec.no_uncaught_exception`	FAIL	no exception escaped execution	10/10	9
`static.no_unbound_names`	FAIL	every referenced name resolves	10/10	6
`results.values_computed`	FAIL	the call-site argument is not a literal	10/10	3
`static.syntax_valid`	FAIL	source compiles before execution is paid for	10/10	3
`env.clean_namespace`	FAIL	execution began with harness globals only	10/10	—
`env.code_identity`	FAIL	executed source hash matches the recorded hash	10/10	—
`exec.completed_within_budget`	FAIL	the process group finished before timeout	10/10	—
`results.contract_not_shadowed`	FAIL	the injected recording API was not redefined	10/10	—
`results.expected_keys_present`	FAIL	every declared key was recorded	10/10	—
`results.values_finite`	FAIL	no NaN or infinity was recorded	10/10	—
`static.no_banned_calls`	FAIL	no exit()/sys.exit() exit-code forgery	10/10	—

`env.seed_recorded`	WARN	seed metadata is present	10/10	—
`exec.no_swallowed_traceback`	WARN	no traceback was printed and ignored	10/10	—
`logs.model_error_signals`	WARN	model scan found residual suspicious lines	not emitted	—
`logs.no_error_signals`	WARN	deterministic scan found no error signal	10/10	—
`results.non_degenerate`	WARN	no exact 0, 1, or chance-level value	10/10	—
`results.single_observation`	WARN	a key was not silently overwritten	1/10	—
`env.provenance`	INFO	runtime, platform, and library versions recorded	10/10	—
`output.untruncated`	INFO	full stdout/stderr size and path recorded	10/10	—
`report.fixes_grounded`	INFO	ungrounded model-written fixes were discarded	not emitted	—

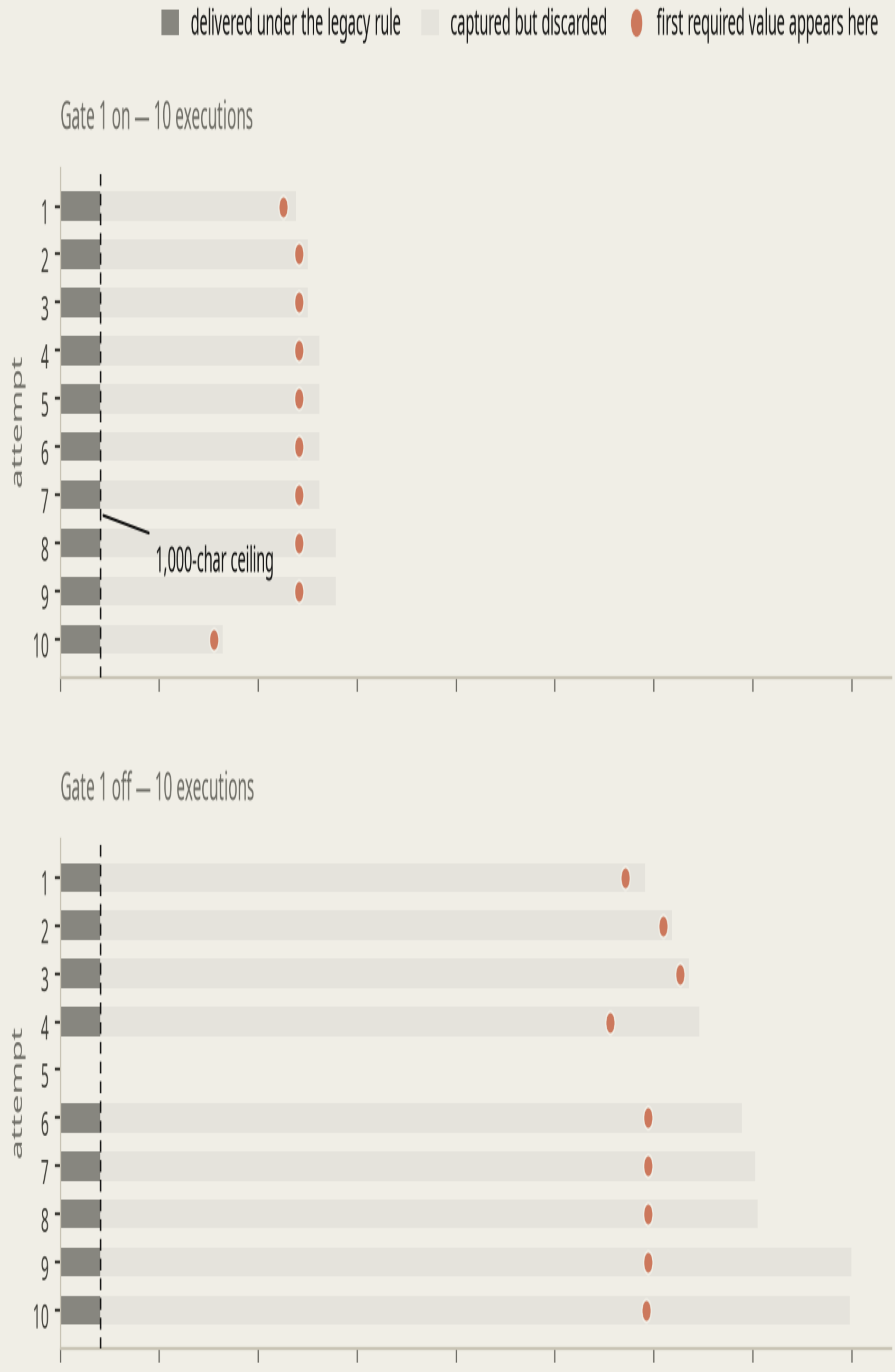
results.single_observation warned on 9/10 gated attempts because prepended data-preparation code and the experiment body both recorded timing. The final best-code re-execution was clean. Warnings never block.

05 Completed gated and ungated workflows

MEASUREMENT	GATE 1 ON	GATE 1 OFF	INTERPRETATION
Accepted executions	10/10	9/10	one ungated syntax failure
Required pairs recorded	40/40	36/40	code-side capture
Required pairs delivered	40/40	0/40	+100 percentage points
Attempts delivering all four	10/10	0/10	+100 percentage points
Required values in final paper	4/4	0/4	registry reached only gated writer
Total stdout captured	62,617	154,596	full logs remain on disk



The causal channel comparison holds gated executions fixed and varies only the evidence route; the independent gate-off arm confirms the mechanism.

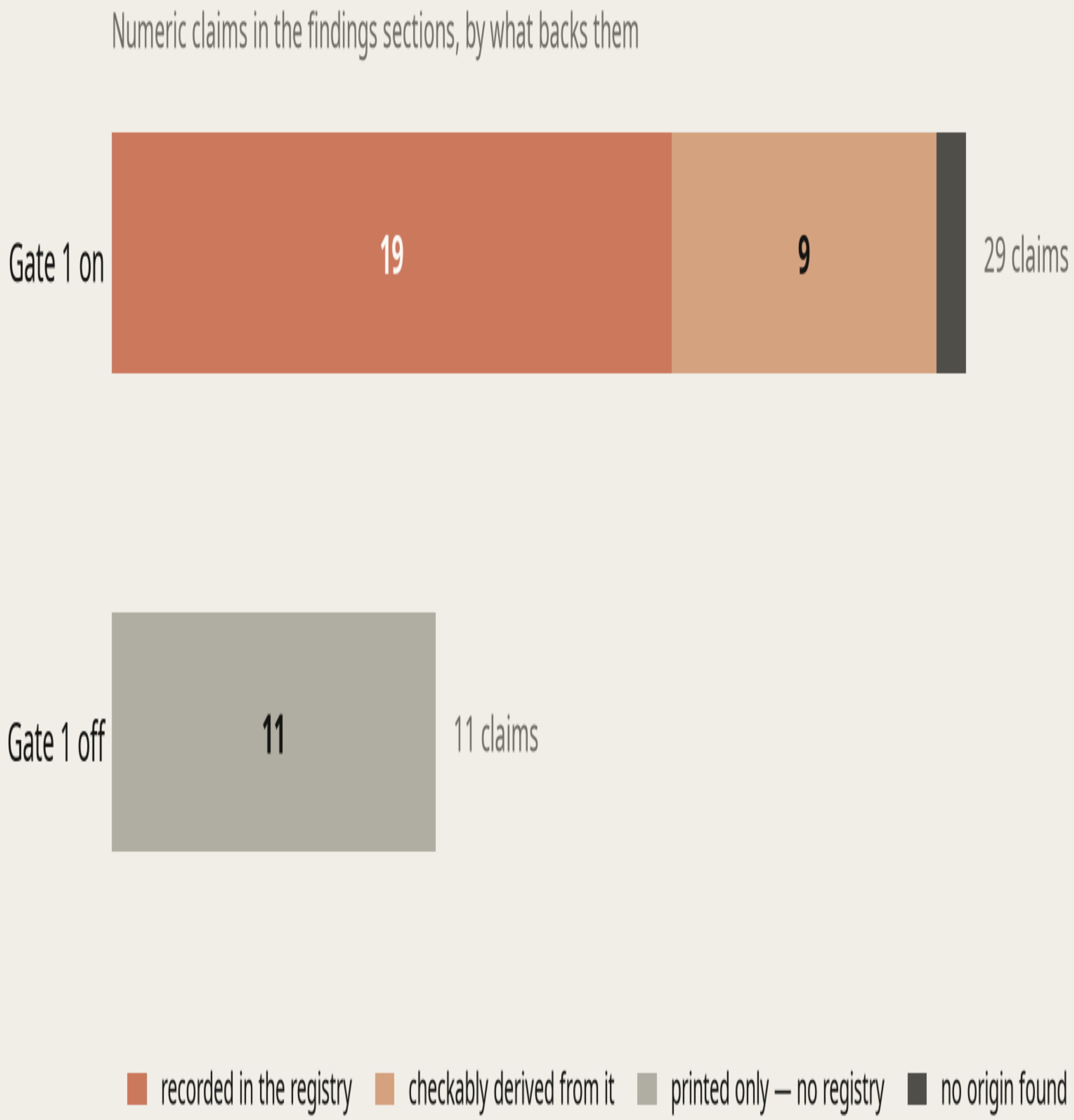


delivered under the legacy rule
captured but discarded
first required value appears here
Gate 1 on – 10 executions
attempt
1
2
3
4
5
6
7
8
9
10
1,000-char ceiling
Gate 1 off – 10 executions
attempt
1
2
3
4
5
6
7
8
9
10

Fixed-artifact counterfactual: 40/40 through Gate 1 versus 0/40 through the legacy view of those exact executions.

06 Generated-paper verification

Both completed workflows produced papers, and both originals are included in this submission. Numeric claims in each findings section were extracted and scored by the same artifact-only procedure.



Claim provenance in the two generated papers. The gated paper's one unmatched item is an external literature comparator, not a run result.

PAPER AUDIT	GATE 1 ON	GATE 1 OFF

Numeric claims extracted	29	11
Direct registry matches	19	0
Checkable registry derivations	9	0
Printed but not registry-bound	0	8
No origin found	1	3
Direct + derived traceability	96.6%	0%
Required four-row metric table	present	absent
Required values reported	4/4	0/4

The ungated paper did not invent the missing required values; the identical honesty instruction caused it to describe the evidence gap. Gate 1 changed delivery and traceability, not the writer's instruction.

BUNDLED ORIGINAL	LOCATION
Gated generated paper	papers/gated/generated_report.txt
Ungated generated paper	papers/ungated/generated_report.txt
Portable LaTeX + figures	included beside each original

07 Accuracy results and causal interpretation

The final papers contain different task accuracies, but those values cannot be treated as a Gate effect: the two stochastic workflows chose different data generators and training schedules. The accurate Gate comparison is evidence completeness on fixed artifacts.

FINAL EXECUTION VALUE	GATE 1 ON	GATE 1 OFF	CAUSAL READING
Accuracy with 100 labels	0.890	0.775	not attributable to Gate 1
Accuracy with 400 labels	0.970	0.810	not attributable to Gate 1
Efficiency ratio	0.9175	0.9568	different sampled experiment

Training seconds	0.0171	0.1070	different schedules/hardware path
Required-result delivery accuracy	40/40 (100%)	0/40 (0%)	paired channel effect
Paper-claim traceability	28/29 (96.6%)	0/11 (0%)	measured provenance effect
Mean reviewer score	3.735	3.765	−0.030; no quality effect measured

Result interpretation. Gate 1 made the measurements auditable and available. It did not make the gated task numerically more accurate, and this one-workflow-per-arm design does not support a scientific-quality treatment estimate.

08 Diagnostic log-scanner benchmark

The only model-assisted component was measured separately on a 68-line labelled corpus containing 34 signals and 34 clean lines. It remains WARN-only regardless of benchmark performance.

SCANNER	PRECISION	RECALL	F1	MISSED SIGNALS
deterministic patterns	1.000	0.529	0.692	16
+ model scan (3-shot)	1.000	0.882	0.938	4

Measured change: recall 0.529 → 0.882 (+35.3 points), F1 0.692 → 0.938, and missed signals 16 → 4 (75% fewer), with zero measured false positives in either configuration.

09 MLR-Bench context

MLR-Bench (arXiv:2505.19955v3) is the closest local benchmark paper because it evaluates open-ended ML experimentation and paper writing. Its task suite and judge rubric differ from this harness, so the comparison is mechanistic context rather than a shared leaderboard.

PUBLISHED BENCHMARK FINDING	VALUE	GATE 1 RELEVANCE
Coding-agent results fabricated or invalidated	80%	partial coverage: execution + provenance
Claude Code tasks using placeholder results	8/10	strong coverage of non-execution and missing provenance

Claude Code completeness / soundness / overall	6.00 / 4.75 / 4.95	complete-looking output can remain unsound
Codex overall	4.95 / 10	below the benchmark's 6.0 threshold
This campaign: required-result delivery	100% vs 0%	directly measured channel mechanism
This campaign: paper traceability	96.6% vs 0%	value-layer provenance, not scientific validity

No MLR-Bench task was run. It would be incorrect to subtract the campaign percentages from MLR-Bench's 80% failure figure. A future study should run at least ten paired benchmark tasks with preregistered validity and provenance measures. The bundled comparator is available at verification/benchmark/MLR_Bench.pdf.

10 Implementation completion and security findings

AREA	COMPLETED VALIDATION OR FIX
Agent Laboratory integration	Expected keys propagate from YAML to Gate1Config; Gate-off accepted/rejected reward paths handle report=None
Feedback rig	Relative retained workdirs resolve correctly; --json emits clean machine-readable output
Credential isolation	Credential-shaped variables are stripped before experiment execution
Linux parent protection	Parent is non-dumpable while child code runs, blocking same-user /proc and ptrace access in regression tests
Campaign launcher	TTY prompt keeps the provider key out of argv and configuration files
Regression suites	286 Gates tests + 76 Agent Laboratory tests = 362 passing

Security boundary. This is process isolation, not a hardened sandbox. Production execution still needs a low-privilege container or account, minimal mounts, and network restrictions. The copied submission artifacts contain no provider credential.

11 Limits and overclaims to avoid

LIMIT	WHAT IT MEANS
One workflow per arm	Ten candidates share workflow history and are not ten independent research tasks.

Computed-value heuristic	A variable assigned from an indirect constant can pass values_computed; it is not proof of measurement.
Presence-only key contract	Expected keys must exist, but extra undeclared metrics do not currently fail.
Phase scope	Prepended data-preparation code can record values that satisfy a later experiment contract.
Task compliance	Gate 1 checks executable validity, not whether every method instruction was followed.
Paper interpretation	Unsupported derivations and scientific overclaims can survive; later report gates remain necessary.
Log scan bound	Full logs are retained, while automated scanning is bounded to two million characters per stream.
Cost	Provider billing was not measured; Agent Laboratory's displayed $0.0 is a placeholder.

Defensible scope: Gate 1 is complete for deterministic execution validity, typed result capture, and causal provenance through the value layer. It is not a standalone anti-fabrication or scientific-soundness system.

12 Professor verification package

The final folder is self-indexing and includes the original papers, canonical completed workflow logs, diagnostic first-run log, manifests, shared configuration, all execution artifacts, feedback-loop evidence, source research reports, the local benchmark paper, and cryptographic checksums.

VERIFICATION ITEM	BUNDLED PATH
Gated workflow log	verification/logs/gated_workflow.log
Completed ungated workflow log	verification/logs/ungated_workflow.log
Initial ungated diagnostic log	verification/logs/ungated_initial_failed_workflow.log
Generated papers + figures	verification/papers/
Twenty completed-arm execution artifacts	verification/execution-artifacts/

Feedback reports and deterministic repeats	verification/evidence/
Manifests and common YAML	verification/run-metadata/
Original Claude and Codex source reports	verification/source-reports/
File hashes	SHA256SUMS.txt

Start with README.md and SUBMISSION_MANIFEST.json. The first ungated workflow generated ten attempts but no paper because of a host-integration bug; the completed retry is the canonical ungated comparison.

13 Final decision

Gate 1 testing and implementation validation are complete for the declared scope.

The measured gain is evidence availability and provenance: 40/40 versus 0/40 required values delivered, 96.6% versus 0% paper-claim traceability, and 9 of 18 adversarial rejected turns caught beyond the legacy rule. Continue with strict result schemas, report-contract enforcement, scientific claim validation, a real sandbox, and paired MLR-Bench tasks.